

Module 25: Advanced SQL Analytics

Advanced SQL Analytics combines SQL techniques such as JOINS, CTEs, Window Functions, CASE Statements, and Aggregations to solve real-world business problems. This module focuses on analytics scenarios commonly used by Data Analysts, Business Analysts, Product Analysts, and Analytics Engineers.



Introduction

In previous modules, you learned:

```
SELECT
WHERE
GROUP BY
HAVING
JOINS
Subqueries
CTEs
Window Functions
```

Now it's time to combine them to answer business questions such as:

- How many customers return every month?
- Which products generate the most revenue?
- What is customer retention?
- Where are users dropping from the sales funnel?
- What are the monthly growth trends?

These are Advanced SQL Analytics problems.



Learning Objectives

After completing this module, you will be able to:

- ✓ Perform Cohort Analysis
- ✓ Perform Retention Analysis
- ✓ Build Funnel Analysis
- ✓ Create Pivot Reports
- ✓ Calculate Business KPIs

- ✓ Solve Real Analytics Case Studies
- ✓ Write Production-Level SQL Queries

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1 Advanced Analytics Overview

Most analytics projects involve:

```
Raw Data
  ↓
Cleaning
  ↓
Transformation
  ↓
Aggregation
  ↓
Insights
```

Common SQL Analytics Workflow

```
SELECT
JOIN
```

CASE
GROUP BY
WINDOW FUNCTIONS

Example Business Question

Top 10 customers by revenue

Solution:

JOIN
GROUP BY
SUM()
RANK()

2 Business KPIs

KPI = Key Performance Indicator

Common KPIs

KPI	Formula
Revenue	SUM(SalesAmount)
Customers	COUNT(DISTINCT CustomerID)
Orders	COUNT(OrderID)
Average Order Value	Revenue / Orders
Retention Rate	Returning Users / Total Users
Conversion Rate	Purchases / Visitors

Revenue KPI

SELECT
SUM(SalesAmount) AS Revenue

```
FROM Sales;
```

Average Order Value

```
SELECT
SUM(SalesAmount) /
COUNT(OrderID) AS AvgOrderValue
FROM Orders;
```

3 Cohort Analysis

One of the most common analytics techniques.

Purpose

Track users based on when they first joined.

Example

Customer	Join Month
C1	Jan
C2	Jan
C3	Feb

Question

```
How many January users returned in February?
```

Cohort Query

```
WITH CustomerCohort AS
(
  SELECT CustomerID,
```

```
MIN(OrderDate) AS FirstOrderDate

FROM Orders

GROUP BY CustomerID
)

SELECT *
FROM CustomerCohort;
```

Business Uses

- Customer Retention
 - Product Adoption
 - Subscription Analysis
-



Retention Analysis

Measures customer return behavior.

Formula

```
Returning Customers
-----
Total Customers
```

Example

```
WITH MonthlyCustomers AS
(
    SELECT
        CustomerID,
        MONTH(OrderDate) AS OrderMonth

    FROM Orders
)

SELECT *
FROM MonthlyCustomers;
```

Uses

- SaaS Products
 - E-commerce
 - Mobile Apps
-

Important KPI

Retention Rate

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Funnel Analysis

Tracks user journey.

Example Funnel

Visit Website
↓
Sign Up
↓
Add Product
↓
Purchase

Example Data

CustomerID	Stage
1	Visit
1	Signup
1	Purchase

Query

```
SELECT Stage,
        COUNT(DISTINCT CustomerID)
FROM FunnelData
GROUP BY Stage;
```

Output

Stage	Users
Visit	1000
Signup	500
Purchase	200

Conversion Rate

Purchase Users

Visit Users

Business Uses

- Product Analytics
- Marketing Analytics
- Sales Optimization



Pivoting Data

Converts rows into columns.

Example

Month	Revenue
Jan	1000
Feb	1500

SQL Server PIVOT

```
SELECT *
FROM Sales
PIVOT
(
    SUM(Revenue)
    FOR Month IN
        (
            [Jan],
            [Feb],
            [Mar]
        )
) P;
```

Alternative Using CASE

```
SELECT

SUM(
CASE
WHEN Month='Jan'
THEN Revenue
ELSE 0
END
) AS JanuaryRevenue,

SUM(
CASE
WHEN Month='Feb'
THEN Revenue
ELSE 0
END
) AS FebruaryRevenue

FROM Sales;
```

Used In

- Dashboards
 - Reports
 - Excel-style Tables
-

7 Customer Segmentation

Divide customers into groups.

Example

Purchase Amount	Segment
> 100000	Premium
> 50000	Gold
Otherwise	Regular

Query

```
SELECT CustomerID,  
  
CASE  
  
WHEN TotalPurchase > 100000  
THEN 'Premium'  
  
WHEN TotalPurchase > 50000  
THEN 'Gold'  
  
ELSE 'Regular'  
  
END AS Segment  
  
FROM Customers;
```

Business Uses

- Marketing Campaigns
 - Customer Targeting
 - Loyalty Programs
-

8 RFM Analysis

Very common interview topic.

RFM

Recency
Frequency
Monetary

Recency

How recently customer purchased.

Frequency

How often customer purchases.

Monetary

How much customer spends.

Query

```
SELECT

CustomerID,

DATEDIFF
(
    DAY,
    MAX(OrderDate),
    GETDATE()
) AS Recency,

COUNT(*) AS Frequency,

SUM(SalesAmount) AS Monetary

FROM Orders

GROUP BY CustomerID;
```

Uses

- Customer Scoring
 - Marketing Segmentation
-



Growth Analysis

Measures business growth.

Example

Monthly Revenue

Month	Revenue
Jan	1000
Feb	1500

Growth

$$1500 - 1000 = 500$$

SQL Query

```
SELECT
  Month,
  Revenue,
  Revenue -
    LAG(Revenue)
  OVER
    (
      ORDER BY Month
    )
  AS Growth
FROM Sales;
```

Percentage Growth

```
SELECT  
  
Month,  
  
(  
Revenue -  
LAG(Revenue)  
OVER  
(  
ORDER BY Month  
)  
)  
  
* 100.0  
  
/  
  
LAG(Revenue)  
OVER  
(  
ORDER BY Month  
)  
  
AS GrowthPercent  
  
FROM Sales;
```

Revenue Analytics

Most common business metric.

Revenue by Product

```
SELECT ProductID,  
       SUM(SalesAmount)  
  
FROM Sales  
  
GROUP BY ProductID;
```

Revenue Ranking

```
SELECT ProductID,  
  
SUM(SalesAmount) AS Revenue,  
  
RANK()  
OVER  
(  
ORDER BY SUM(SalesAmount) DESC  
)  
  
AS RevenueRank  
  
FROM Sales  
  
GROUP BY ProductID;
```

Revenue Contribution %

```
SELECT ProductID,  
  
SUM(SalesAmount) * 100.0  
  
/  
  
SUM(SUM(SalesAmount))  
OVER()  
  
AS ContributionPercent  
  
FROM Sales  
  
GROUP BY ProductID;
```

Top-N Analysis

Frequently asked in interviews.

Top 5 Customers

```
WITH CustomerRevenue AS
(
    SELECT CustomerID,

        SUM(SalesAmount) AS Revenue

    FROM Sales

    GROUP BY CustomerID
)

SELECT *

FROM
(
    SELECT *,

        ROW_NUMBER()
        OVER
        (
            ORDER BY Revenue DESC
        ) AS RN

    FROM CustomerRevenue
) A

WHERE RN <= 5;
```

Bottom 5 Products

Same logic using ascending order.

1 2 Advanced Dashboard Queries

Real dashboard metrics often combine:

```
CTEs
Window Functions
CASE
GROUP BY
```

Example

```
WITH MonthlyRevenue AS
(
    SELECT
        MONTH(OrderDate) AS SalesMonth,

        SUM(SalesAmount) AS Revenue

    FROM Sales

    GROUP BY MONTH(OrderDate)
)

SELECT

SalesMonth,
Revenue,

Revenue -

LAG(Revenue)
OVER
(
    ORDER BY SalesMonth
)

AS MonthlyGrowth

FROM MonthlyRevenue;
```

Dashboard Output

Month	Revenue	Growth
Jan	1000	NULL
Feb	1500	500

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Real Interview Case Studies

Case Study 1

Find Top 3 Highest-Paying Departments.

Skills:

```
GROUP BY  
SUM  
RANK
```

Case Study 2

Find Customers Who Never Ordered.

Skills:

```
LEFT JOIN  
IS NULL
```

Case Study 3

Calculate Running Revenue.

Skills:

```
SUM OVER
```

Case Study 4

Find Second Highest Salary.

Skills:

```
DENSE_RANK
```

Case Study 5

Month-over-Month Growth.

Skills:

LAG

Best Practices

Use CTEs for Readability

Use Window Functions for Analytics

Avoid Deep Nested Queries

Use Meaningful Aliases

Verify KPI Logic

Business definitions matter.

Test Edge Cases

NULL values.

Missing months.

Duplicate rows.

SQL Analytics Toolkit

A strong analyst should master:

```
JOINS
GROUP BY
HAVING
CASE
CTEs
Window Functions
Subqueries
```



Summary

In this module, you learned:

- ✓ Cohort Analysis
 - ✓ Retention Analysis
 - ✓ Funnel Analysis
 - ✓ Pivot Reports
 - ✓ Customer Segmentation
 - ✓ RFM Analysis
 - ✓ Growth Analysis
 - ✓ Revenue Analytics
 - ✓ Top-N Analysis
 - ✓ Dashboard Queries
 - ✓ Interview Case Studies
-



Complete SQL Roadmap Summary

You have now completed:

SQL Fundamentals

```
SELECT
WHERE
ORDER BY
INSERT
UPDATE
DELETE
```

Database Design

```
Tables
Constraints
Keys
ALTER TABLE
```

Data Retrieval

JOINS
Subqueries
Views

Programming

Stored Procedures
Functions

Performance

Indexes

Analytics

GROUP BY
HAVING
CASE
CTEs
Window Functions
Advanced Analytics



Final Practice Projects

Project 1

HR Analytics Dashboard

Features:

- Employee KPIs
 - Department Rankings
 - Salary Analysis
 - Attrition Metrics
-

Project 2

E-Commerce Analytics

Features:

- Revenue Analysis
 - Customer Segmentation
 - Product Rankings
 - Retention Analysis
-

Project 3

Sales Analytics Dashboard

Features:

- Running Revenue
 - Growth Analysis
 - Top Products
 - Regional Performance
-

Project 4

Customer Analytics

Features:

- RFM Analysis
 - Cohort Analysis
 - Funnel Analysis
 - Customer Lifetime Value
-



Final Repository Structure

```
SQL-Zero-to-Hero/  
|  
├─ Module_01_SQL_Fundamentals/  
├─ Module_02_Database_Objects/  
├─ Module_03_SQL_Data_Types/  
├─ Module_04_Create_Table/  
├─ Module_05_Constraints/  
├─ Module_06_Insert/  
├─ Module_07_Select/  
├─ Module_08_Where/  
├─ Module_09_Order_By/  
└─ Module_10_Aggregates/
```

```
|— Module_11_Group_By/
|— Module_12_Having/
|— Module_13_Update/
|— Module_14_Delete/
|— Module_15_Alter_Table/
|— Module_16_Joins/
|— Module_17_Subqueries/
|— Module_18_Case_Statement/
|— Module_19_Views/
|— Module_20_Indexes/
|— Module_21_Stored_Procedures/
|— Module_22_Functions/
|— Module_23_CTEs/
|— Module_24_Window_Functions/
|— Module_25_Advanced_SQL_Analytics/
|
|— Projects/
|   |— HR_Analytics/
|   |— Sales_Analytics/
|   |— Customer_Analytics/
|   |— Ecommerce_Analytics/
|
|— Datasets/
|— Interview_Questions/
|— CheatSheets/
|— README.md
```